

UNIT 7

EMBEDDED SYSTEMS – AN OVERVIEW

Embedded systems are a cornerstone of the electronics industry today. Embedded systems are used in almost all the areas like consumer, cooking, industrial, automotive, medical, commercial and military applications.

What is a system?

A system is a way of working, organizing or doing one or many tasks according to a fixed plan, program or set of rules. A system is also an arrangement in which all its units assemble and work together according to the plan or program.

Embedded System

As its name suggests, “embedded” means something that is attached to another thing. An embedded system can be thought of as a computer hardware system having software embedded in it. An embedded system can be an independent system, or it can be a part of a large system. It offers many benefits such as sophisticated control, precision timing, low unit cost, low development cost, high flexibility, small size, and low weight. These basic characteristics can be used to improve the overall system or device in various ways:

- Improved performance
- More functions and features
- Reduced cost
- Increased dependability

Because of these benefits, billions of microcontrollers are sold each year to create embedded systems for a wide range of products. In other words: An embedded system is a microcontroller or microprocessor-based system which is designed to perform a specific task. For example, a fire alarm is an embedded system; it will sense only smoke.

Embedded systems contain two main elements:

- Embedded system hardware: As with any electronic system, an embedded system requires a hardware platform on which to run. The hardware will be based around a microprocessor or microcontroller. The embedded system hardware will also contain other elements including memory, input output (I/O) interfaces as well as the user interface, and the display.
- Embedded system software: The embedded system software is written to perform a particular function. It is typically written in a high-level format and then compiled

down to provide code that can be lodged within a non-volatile memory within the hardware.

The various embedded system blocks are briefly explained below:

- Sensor – It measures the physical quantity and converts it to an electrical signal which can be read by an observer or by any electronic instrument like an A2D converter. A sensor stores the measured quantity to the memory.
- A-D Converter – An analog-to-digital converter converts the analog signal sent by the sensor into a digital signal.
- Processor – Processors process the data to measure the output and store it to the memory.
- D-A Converter – A digital-to-analog converter converts the digital data fed by the processor to analog data.
- Actuator – An actuator compares the output given by the D-A converter to the actual (expected) output stored in it and stores the approved output.

Embedded System Hardware

When using an embedded system there is a choice between the use of a microcontroller or a microprocessor.

- Microcontroller based systems: A microcontroller is essentially a CPU, central processing unit, or processor with integrated memory or peripheral devices. As fewer external components are needed, embedded systems using microcontrollers tend to be more widely used.
- Microprocessor based systems: Microprocessors contain a CPU but use external chips for memory and peripheral interfaces. As they require more devices on the board, but they allow more expansion and selection of exact peripherals etc., this approach tends to be used for the larger embedded systems.

Embedded System Software

One of the key elements of any embedded system is the software that is used to run the microcontroller.

There is a variety of ways that this can be written:

- Machine code: Machine code is the most basic code that is used for the processor unit. The code is normally in hex code and provides the basic instructions for each operation of the processor. This form of code is rarely used for embedded systems these days.
- Programming language: Writing machine code is very laborious and time consuming. It is difficult to understand and debug. To overcome this, high level programming languages are often used, including C, C++, etc.

Embedded System Architecture

There are two basic types of embedded system architecture. When data and code lie in different memory blocks, then the architecture is referred to as Harvard architecture. In case data and code lie in the same memory block, then the architecture is referred to as Von Neumann architecture.

Von Neumann Architecture

The Von Neumann architecture was first proposed by a computer scientist John von Neumann. In this architecture, one data path or bus exists for both instruction and data. As a result, the CPU does one operation at a time. It either fetches an instruction from memory or performs read/write operation on data. So, an instruction fetch and a data operation cannot occur simultaneously, sharing a common bus.

Von-Neumann architecture supports simple hardware. It allows the use of a single, sequential memory. Today's processing speeds vastly outpace memory access times, and we employ a very fast but small amount of memory (cache) local to the processor.

Harvard Architecture

The Harvard architecture offers separate storage and signal buses for instructions and data. This architecture has data storage entirely contained within the CPU, and there is no access to the instruction storage as data. Computers have separate memory areas for program instructions and data using internal data buses, allowing simultaneous access to both instructions and data.

Programs needed to be loaded by an operator; the processor could not boot itself. In Harvard architecture, there is no need to make the two memories share properties.

Conclusion

Many embedded systems have requirements that differ significantly both in their details and scope from desktop computers. In particular, the demands of the specific application and the interface with external equipment may dominate the system design. Also, long life-cycles and in some cases extreme cost sensitivity require more attention to optimization based on these goals rather than maximizing the computational throughput.

The business and cultural climates in many embedded system design situations are such that traditional simulation-based computer design techniques may not be viable in their current form. Such methodologies may not be cost-effective given constraints on categories of expenditures, may not be seen as worthwhile by non-computer-trained professionals, or may simply be solving the wrong problems.

Recent interest in hardware/software code design is a step in the right direction, as it permits trade-offs between hardware and software that are critical for more cost-effective embedded systems. However, to be successful future tools may well need to increase scope even further to include life-cycle issues and business issues.

Exercises

1. Answer the following questions.

- a) What is the general position of embedded systems in the electronics industry today?
- b) Write down the two definitions of a system.
- c) What does the term “embedded” mean?
- d) How can an embedded system be thought of and in which two ways can it occur?
- e) Which benefits are offered by an embedded system?
- f) What are the basic characteristics (benefits) of an embedded system?
- g) Give the definition and an example of an embedded system.
- h) What do you know about the embedded system hardware?
- i) What do you know about the embedded system software?
- j) What is the role of a sensor in an embedded system?
- k) What is the role of an A-D converter in an embedded system?
- l) What is the role of a processor in an embedded system?
- m) What is the role of a D-A converter in an embedded system?
- n) What is the role of an actuator converter in an embedded system?
- o) What do you know about microcontroller based embedded systems?
- p) What do you know about microprocessor based embedded systems?
- q) What do you know about the use of machine code in writing the software that is used to run the microcontroller?
- r) What do you know about the use of programming languages in writing the software that is used to run the microcontroller?
- s) What is the basic difference between the Harvard architecture and Von Neumann architecture of embedded systems?
- t) What do you know about the Von Neumann architecture of embedded systems?
- u) What do you know about the Harvard architecture of embedded systems?
- v) In which ways do the requirements of many embedded systems differ significantly both in their details and scope from desktop computers?
- w) Describe the features of the e business and cultural climates in many embedded system design situations?
- x) In terms of the development of embedded systems, why is the recent interest in hardware/software code-design a step in the right direction?
- y) What should be done with embedded systems in order to make them successful future tools?

2. Explain/describe the following terms and expressions, on the basis of their use in the texts. Use a dictionary or check online if necessary.

cornerstone	
assemble	
sophisticated	
dependability	
hardware platform	
microprocessor	
microcontroller	
lodged	
non-volatile memory	
compiled down	
external chips	
peripheral interfaces	
machine code	
memory block	
outpace	
life-cycle	
computational throughput	
constraints	
given	
trade-off	
digital-to-analog converter	
hex code	

3. Write the descriptions of the modified nouns from the table.

computer hardware	system	
microprocessor based	system	
embedded system	hardware	
input output	interfaces	
embedded system	software	
embedded system	blocks	
high level programming	languages	
embedded system	architecture	
memory access	times	
separate memory	areas	
internal data	buses	
embedded system design	situations	
traditional simulation-based computer design	techniques	
non-computer-trained	professionals	

hardware/software code	design	
cost-effective embedded	systems	

4. Connect the sentence parts in accordance with the text and their order to get correct sentences.

A system is a way of working, organizing or doing one or many tasks	to measure the output and store it to the memory.
A system is an arrangement in which all its units assemble	which can be read by an observer or by any electronic instrument.
An embedded system can be thought of as	the digital data fed by the processor to analog data.
An embedded system is a microcontroller or microprocessor based system	the analog signal sent by the sensor into a digital signal.
As with any electronic system, an embedded system	then the architecture is referred as Von Neumann architecture.
The embedded system software is typically written in a high level format	the basic instructions for each operation of the processor.
Sensor measures the physical quantity and converts it to an electrical signal	then the architecture is referred as Harvard architecture.
An analog-to-digital converter converts	the software that is used to run the microcontroller.
Processors process the data	or processor with integrated memory or peripheral devices.
A digital-to-analog converter converts	a computer hardware system having software embedded in it.
An actuator compares the output given by the D-A converter	according to a fixed plan, program or set of rules.
A microcontroller is essentially a CPU, central processing unit,	and work together according to the plan or program.
Microprocessors contain a CPU but use external chips	and then compiled down to provide code that can be lodged within a non-volatile memory within the hardware.
One of the key elements of any embedded system is	which is designed to perform a specific task.
Machine code is normally in hex code and provides	requires a hardware platform on which to run.

When data and code lie in different memory blocks,	to the actual (expected) output stored in it and stores the approved output.
In case data and code lie in the same memory block,	for memory and peripheral interfaces.

5. Complete the text with the given words and expressions.

| interfaces | electrical | performance | electronic | networks | processor |
microcontrollers | microprocessors | portable devices | programmable | chips |
physical | input/output | processor | computations |

An embedded system is a computer system—a combination of a computer_____, computer memory, and _____ peripheral devices—that has a dedicated function within a larger mechanical or _____system. It is embedded as part of a complete device often including electrical or _____hardware and mechanical parts. Because an embedded system typically controls _____operations of the machine that it is embedded within, it often has real-time computing constraints. Embedded systems control many devices in common use today. Ninety-eight percent of all _____manufactured are used in embedded systems.

Modern embedded systems are often based on _____ (i.e. microprocessors with integrated memory and peripheral_____), but ordinary microprocessors (using external _____for memory and peripheral interface circuits) are also common, especially in more complex systems. In either case, the processor(s) used may be types ranging from general purpose to those specialized in a certain class of_____, or even custom designed for the application at hand. A common standard class of dedicated processors is the digital signal _____ (DSP).

Since the embedded system is dedicated to specific tasks, design engineers can optimize it to reduce the size and cost of the product and increase the reliability and_____. Some embedded systems are mass-produced, benefiting from economies of scale.

Embedded systems range from_____ such as digital watches and MP3 players, to large stationary installations like traffic light controllers, _____logic controllers, and large complex systems like hybrid vehicles, medical imaging systems, and avionics. Complexity varies from low, with a single microcontroller chip, to very high with multiple units, peripherals and _____mounted inside a large equipment rack.

UNIT 8

INTRODUCTION TO EMBEDDED SYSTEMS

Embedded Systems

The number of embedded systems, also known as integrated systems, in the world is approximately 10,000 million, a figure that keeps increasing. Embedded systems are not only used for industrial purposes but they are present in every aspect of our lives, since a large amount of daily equipment are provided with these systems, such as cars, lifts, toys, etc.

Embedded systems are thought to be closed systems without interfaces with the user or with other devices. However, they are fully functional systems the purpose of which is to execute control tasks. It is also widely believed that all embedded systems are real-time systems but they do need such characteristics by default. Basically, an embedded system has two main features:

Reliability: They must be able to operate continuously without supervision.

Customisation: They must allow the modification of the system based on the needs of the process.

The main problem of embedded systems is that, in general terms, they are designed without taking into account security, offering limitless connectivity. Besides, the protection measures the market offers are very limited.

Embedded operating systems

The first embedded operating systems were exclusively based on the processor (micro-controller), without an operating system and with instructions directly written on it. When the processor was obsolete, so was the device and thus, the programming.

In the 80s, the first embedded systems with operating systems were created mainly to cover communication needs, making hardware applications independent and allowing for the reuse of libraries. The characteristics of modern operating systems include (even though not all embedded systems use them):

- Graphical user interface;
- Implementation of protocol stacks;
- Support for external storage such as USB.

General purpose operating systems have also accessed embedded systems and with them, the weaknesses they involve. Therefore, the weaknesses of external libraries or own libraries of Linux and Windows may cause problems for the devices they control. Can you picture an embedded system blocked by a blue window of

Windows? It may happen but it is more common that the display is independent from control so, even though information may not be shown on the screen, the process is still being executed properly.

Main security problems

As in most industrial equipment, embedded systems are designed to be secure at a physical level, having rebooting measures in the event of failure (watchdog), duplicated hardware parts, fail-proof programming, etc. But, what about cybersecurity?

Some of the weaknesses of the embedded systems affect cybersecurity at a logic level are the following:

Secret sharing: Certain sensitive information, such as passwords, are often shared between the applications of the manufacturer so that they can communicate more easily to one another. This shared information may allow a potential attacker access with the same username to different applications.

Private certificates: Many services require self-signed certificates which are not backed-up by trusted entities, lowering the security level of services. An attacker may use these certificates by using other ones which have been generated specifically and allowing them to perform actions in the device or deceive the user.

Embedded passwords and back doors: These two characteristics are very common. The manufacturer includes embedded passwords preventing their modification over time and making all clients use the same ones. Back doors allow manufacturers to access embedded devices without taking into account any possible changes by the client. Attackers may take advantage of these weaknesses to retrieve information from the device or even to control it.

Free code failures: Most parts of the code of embedded systems are based on routines or reused processes, many of which are based on free codes. This code is not usually checked and it may cause failures with other parts of the system or due to programming weaknesses. Attackers may use these failures as access points for more advanced attacks.

Weak or non-existent cryptography: Encryption methods are usually scarce. The use of encrypted communications is only used in the more advanced versions, whereas the simplest versions work with Telnet and HTTP communications, with the corresponding weaknesses. This limitation may allow potential attackers to capture and interpret the traffic exchanged between both devices and they may even retrieve sensitive information from the device.

Weak or non-existent authentication: Authentication of devices is usually based on a PIN code (such as mobile phones) or has no access control at all. Potential attackers may use brute force methods without much difficulty to gain access to devices or applications.

Exploitation of the server software (HMI, management, web): Some of the services offered are based on the client-server mode where the embedded device acts as server. These services may be vulnerable to XSS, SQL injection attacks, etc. If attackers can make use of these weaknesses, service to the device may be rejected and remote codes may not be executed, with the corresponding problems for the operator.

Input/output communications: Outward communication ports are the weakest point of the embedded systems. In general terms, their use may not be disabled so all ports available are always available even though they are not used at all. Ports which are not being used due to the needs of the device may be used by potential attackers as access points to the device to gain control over it.

Ladder logic exploitation: The programming of the control loop is not properly protected most of the time so it may be modified to perform actions other than the ones developed by the manufacturer. Potential attackers may obtain and modify the programming of the device, forcing it to perform different actions and threatening the whole process, even human lives and the environment.

DoS and DDoS: They are the most common consequence before any attack. These devices are not usually provided with a network frame control or equivalent system, so a repeated information request may saturate communication channels. However, communications are not the only aspect that may cause disruptions. There are other design failures within applications, such as the lack of normalisation of input user data, that may cause rejections in the service.

Potential physical options must be added to all these logic weaknesses, since the opening of the device for its analysis and subsequent attacks are always present. These embedded systems are not usually provided with opening detection mechanisms or test port removal used in the design phase of the hardware.

There are many reasons for applying security measures to embedded systems. Manufacturers are responsible for the application of such measures, particularly regarding physical aspects. However, users can also improve security.

Exercises

1. Answer the following questions.

- a) What is the approximate number of embedded systems in the world, and what are they used for?
- b) Give some common thoughts/beliefs and facts about embedded systems.
- c) Name and describe the two main features of an embedded system.
- d) What are the main problems of embedded systems?
- e) What were the first embedded operating systems like?
- f) What were the features of the embedded systems created in the 80s?
- g) What do the characteristics of modern operating systems include?
- h) What have general purpose operating systems accessed together with access to embedded systems?
- i) What may the weaknesses of external libraries or own libraries of Linux and Windows cause? Give an example and discuss it.
- j) In which ways are embedded systems designed, in terms of security?
- k) How does secret sharing present the weakness of the embedded systems in relation to cybersecurity?
- l) How do private certificates present the weaknesses of the embedded systems in relation to cybersecurity?
- m) How do the embedded passwords and back doors present the weaknesses of the embedded systems in relation to cybersecurity?
- n) How do the free code failures present the weaknesses of the embedded systems in relation to cybersecurity?
- o) How does the weak or non-existent cryptography present the weaknesses of the embedded systems in relation to cybersecurity?
- p) How does the weak or non-existent authentication present the weaknesses of the embedded systems in relation to cybersecurity?
- q) How does the weak or non-existent authentication present the weaknesses of the embedded systems in relation to cybersecurity?
- r) How does the exploitation of the server software present the weaknesses of the embedded systems in relation to cybersecurity?
- s) How do input/output communications present the weaknesses of the embedded systems in relation to cybersecurity?
- t) How does the ladder logic exploitation present the weaknesses of the embedded systems in relation to cybersecurity?
- u) How do the DoS and DDoS present the weaknesses of the embedded systems in relation to cybersecurity?
- v) What sorts of weaknesses must be added to the logic weaknesses of embedded systems and why?

w) Which parties are responsible for applying security measures to embedded systems?

2. Explain/describe the following terms and expressions, on the basis of their use in the texts. Use a dictionary or check online if necessary.

approximately	
customisation	
obsolete	
protocol stacks	
external libraries	
cybersecurity	
logic level	
by trusted entities	
cryptography	
encryption	
control loop	
saturate	
disruption	
normalisation	

3. Write the descriptions of the modified nouns from the table.

embedded operating	systems	
general purpose operating	systems	
fail-proof	programming	
self-signed	certificates	
non-existent	cryptography	
brute force	methods	
client-server	mode	
ladder logic	exploitation	
network frame	control	
opening detection	mechanisms	
input user	data	
test port	removal	

4. Match the words on the basis of their use in the text to get proper collocations.

execute	information
write	service
execute	control
design	actions
share	embedded systems

access	control tasks
require	security measures
perform	failure
retrieve	instructions
cause	access
allow	rejection
exchange	to different applications
gain	remote codes
make use of	communication channels
execute	use
reject	process
disable	to capture and interpret
use	traffic
gain	security
modify	certificate
threaten	information
saturate	ports
cause	programming
apply	process
improve	weaknesses

5. Match the cybersecurity weaknesses and their descriptions.

Secret sharing	Encryption methods are usually scarce. The encrypted communications are only used in the more advanced versions, whereas the simplest versions work with Telnet and HTTP communications, with the corresponding weaknesses. This limitation may allow potential attackers to capture and interpret the traffic exchanged between both devices and they may even retrieve sensitive information from the device.
Private certificates	Authentication of devices is usually based on a PIN code (such as mobile phones) or has no access control at all. Potential attackers may use brute force methods without much difficulty to gain access to devices or applications
Embedded passwords and back doors These	Certain sensitive information, such as passwords, are often shared between the applications of the manufacturer so that they can communicate more easily to one another. This shared information may allow a potential attacker access with a same username to different applications
Free code failures	Many services require self-signed certificates which are not backed-up by trusted entities, lowering the security level of services. An attacker may use these certificates by

	using other ones which have been generated specifically and allowing them to perform actions in the device or deceive the user.
Weak or non-existent cryptography	The manufacturer includes embedded passwords preventing their modification over time and making all clients use the same ones. Back doors allow manufacturers to access embedded devices without taking into account any possible changes by the client. Attackers may take advantage of these weaknesses to retrieve information from the device or even to control it.
Weak or non-existent authentication	Most parts of the code of embedded systems are based on routines or reused processed, many of which are based on free codes. This code is not usually checked and it may cause failures with other parts of the system or due to programming weaknesses. Attackers may use these failures as access points for more advanced attacks.
Exploitation of the server software	The programming of the control loop is not properly protected most of the times so it may be modified to perform actions other than the ones developed by the manufacturer. Potential attackers may obtain and modify the programming of the device, forcing it to perform different actions and threatening the whole process, even human lives and the environment.
Input/output communications	They are the most common consequence before any attack. These devices are not usually provided with a network frame control or equivalent system, so a repeated information request may saturate communication channels. However, communications are not the only aspect that may cause disruptions. There are other design failures within applications, such as the lack of normalisation of input user data, that may cause rejections in the service.
Ladder logic exploitation	Some of the services offered are based on the client-server mode where the embedded device acts as server. These services may be vulnerable to XSS, SQL injection attacks, etc. If attackers can make use of these weaknesses, service to the device may be rejected and remote codes may not be executed, with the corresponding problems for the operator.
DoS and DDoS	Outward communication ports are the weakest point of the embedded systems. In general terms, their use may not be disabled so all ports available are always available even though they are not used at all. Ports which are not

	being used due to the needs of the device may be used by potential attackers as access points to the device to gain control over it.
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6. Complete the text with the given words and expressions.

| networking | microwave | deployment | systems | Industrial | routers |
 monitoring | wireless | software | embedded | electronic | microcontrollers |
 Operating Systems | developers | consumption |

Embedded _____ are part and parcel of every modern _____ component. These are low power _____ units that are used to run specific tasks for example remote controls, washing machines, _____ ovens, RFID tags, sensors, actuators and thermostats used in various applications, _____ hardware such as switches, _____, modems, mobile phones, PDAs, etc.

Usually, embedded devices are a part of a larger device where they perform specific tasks of the device. For example, embedded systems are used as networked thermostats in Heating, Ventilation and Air Conditioning (HVAC) systems, in Home Automation embedded systems are used as wired or _____ networking to automate and control lights, security, audio/visual systems, sense climate change, _____, etc.

Embedded systems will also be at the cornerstone for the _____ of many Internet of Things (IoT) solutions, especially within certain industry verticals and _____ Internet of Things (IIoT) applications.

Major players in embedded system hardware and, _____ developments are aiming to bring these transformations into their products to take advantage of the growing IoT market. The areas that are going to transform are Real Time _____ (RTOS) and _____ microprocessors and _____, followed by memory footprints and networking, open source communities and _____.

The markets for overall _____ systems will grow with a CAGR of 22.5% to reach \$226 billion through 2020s.